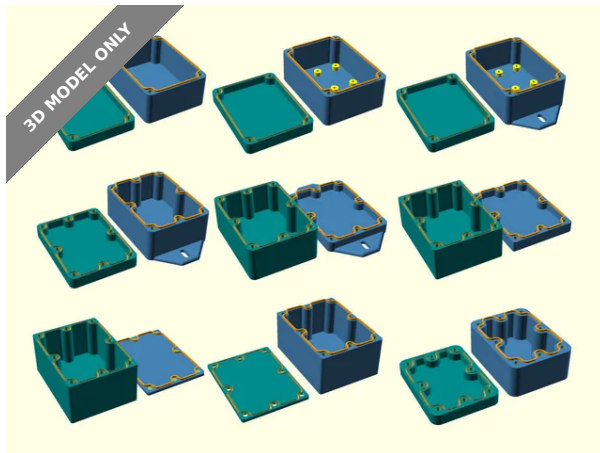




Customizable parametric stable and waterproof electronics ip65 ip67 project box enclosure case housing



[VIEW IN BROWSER](#)

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Summary

Highly scalable waterproof electronic case for 3D printing. ip65 / ip67 Try it out!

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Tags: [parametric](#) [box](#) [nut](#) [case](#) [enclosure](#) [arduino](#)
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Stable and waterproof OpenSCAD case V7.4 by pbtec

See here a wonderful comment from a user:

"This is INSANE! You sir deserve deserve the biggest applaud out there. This is a box to rule them all! ... I have to admit I first didn't understand what you meant with SCAD possibilities, and resized it on my own by cutting xD **then I checked out SCAD version and my jaw simply dropped to the ground!** ... This is a GREAT project!"

Highly scalable electronic case for 3D printing. Try it out!

You'll need to download the free, opensource OpenSCAD software from <https://openscad.org/> and install it. Once installed, open the ".scad" file in Openscad. Activate Customizer under View/Customizer and have fun experimenting!

Fixed version 7.4

It seems that special characters in the code have again led to the code not working properly for some. I apologize for this. **I have now created a new version 7.4** where this is fixed. I would be very interested to know if everything works as expected for you now. Thanks for your feedback in the comments

Enhancements in Version 7.2:

- **Hot melt copper nut support**
- **Screw hole options:** Can be configured to pass through the entire case, with the option to position the nut at the bottom of the case
- **Flexible mounting options:** 1 to 3 holes on all sides for cable glands or switches
- **Improved PCB support:** Now compatible with up to 3 different PCBs holders instead of just one
- **Base bottom mounting holes added** (not recommended if the housing should be highly waterproof)
- **Gasket can now be printed** (TPU).
- **Expanded screw support:** In addition to countersunk screws, round, flat, and hex head screws are now supported

I would love to hear whether you like the new version!

No support needed to print

You don't know Openscad? No problem. Customize the case in Openscad and then export the stl file from Openscad (F6 then F7). Then Import the stl file into your best known 3d program to make holes for example. You can also make the holes afterwards with the drilling machine ;-)

Additional Screws (M2,M2.5,M3,M4 or M5), the matching nuts (hex or square) needed.

If you want to protect your electronics from humidity also "silicone sealing cord" is needed. Available at AliExpress or ebay.

For waterproof cases you can use silicone sealing cord with diameters from 1 to 3mm

To use also without sealing cord. The groove and ridge gives the housing a high stability and tightness.

Use of regular nuts or square nuts

Define outer vertical radius of corners

Echo output in console shows inner and outer size and more

Echo output shows the needed length of the screws

Use screws from m2 up to m5

Default are 4 screws at each corner. For large cases add additional ones in the middle of both x and y sides if needed

Use several predefined wall mount holder, some with multiple mounting holes (up to 3)

Important!

If you use standard or square nuts, you need to pause the printer a certain level to insert the nuts

If you need a stable and waterproof case please print with 100% infill

I'm aware that sometimes, when using too big or too small parameters, there are some rendering issues. To prevent such issues change only one parameter at once and check the result.

I never tested the water proofness of this case. For more information see https://blog.prusaprinters.org/watertight-3d-printing-part-2_53638/

pbtec / pb-tec.ch

<https://www.instagram.com/pbtecch/>

For Projects with ESP32, ESP8266, Arduino, Raspberry, Sonoff, Shelly, IoBroker, IoT. In- and outdoor :-)

Let me see your prints, and remakes and pls don't forget to like this print

use <https://paypal.me/pbtec> if you want to spend me a cold beer. Thanks in advance :-)

If you want to support my project, please use the links below to start your shopping. No additional charges for you, but I get a commission of your purchases

Brass hot melting insert nuts (starter) sets:

https://s.click.aliexpress.com/e/_DezuAlH

https://s.click.aliexpress.com/e/_DeWsPMp

Or individual sizes:

https://s.click.aliexpress.com/e/_DkaQObj

https://s.click.aliexpress.com/e/_DCrO22p

I also recommend using special hot iron to insert the nuts in the right angle:

https://s.click.aliexpress.com/e/_DcHd8Ep

round silicone sealing cord **foam white**

Aliexpress: https://s.click.aliexpress.com/e/_AdVXC6

Ebay: [https://www.ebay.com/itm/184394411044?](https://www.ebay.com/itm/184394411044?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

[mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to](https://www.ebay.com/itm/184394411044?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

round silicone sealing cord **foam black**

Aliexpress: https://s.click.aliexpress.com/e/_A32aUQ

Ebay: [https://www.ebay.com/itm/353309685702?](https://www.ebay.com/itm/353309685702?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

[mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to](https://www.ebay.com/itm/353309685702?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

round silicone sealing **solid** cord **white**

Aliexpress: https://s.click.aliexpress.com/e/_A07nM2

Ebay: [https://www.ebay.com/itm/133527877200?](https://www.ebay.com/itm/133527877200?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

[mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to](https://www.ebay.com/itm/133527877200?mkcid=1&mkrid=711-53200-19255-0&siteid=0&campid=5338823684&customid=&to)

Some tags to better find this project:

Electrical Junction, Distribution box, Connection box, Cable Branch box, Electronic, Housing, ip67, DIY, Project Box, Instrument Case, Wiring socket box, Sealed, Cable gland, Shell, weatherproof, outdoor, airtight, watertight, dustproof, electronics, pcb, airproof, enclosure, parametric

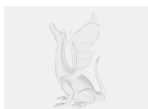
Model files



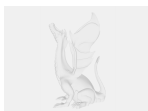
sample2-ultimatewaterproofcase_v6.stl



sample1-ultimatewaterproofcase_v6.stl



ultimatewaterproofcase_v61.scad



ultimatewaterproofcase_v74.scad

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